Which Test Next?

Budget-Aware Antimalarial Hit Confirmation

A seven-week computational research proposal | Revised 27 September 2026

Project in plain language

A promising molecule usually needs several checks. Testing everything is expensive; always running the same test first may waste effort. This project asks whether a learning system can choose **which existing compound to check next, and which check to run first**, so that more promising compounds are confirmed within the same testing budget.

The study uses public experimental results as a hidden answer sheet. The computer must spend a simulated budget to reveal each result. No new experiments or generated molecules are needed to evaluate its decisions.

Exact research question

Can learning from partially observed assay results, combined with adaptive compound-and-assay selection, confirm more TCAMS compounds meeting predefined measured criteria than the best fixed test order and compound-only active learning, at the same assay budget?

What changes from the previous proposal

The project replaces molecular generation and parent-to-analogue change prediction with an experimentally grounded decision benchmark. Molecular fingerprints, lightweight prediction models, uncertainty checks and chemical-family separation remain. The new central object is an action: **(compound, next test)**.

Research contribution to test

Active learning in malaria, generative antimalarial design, multistage prediction and choosing separate constraint evaluations already have substantial precedents. [2-10] The candidate contribution is a reproducible TCAMS benchmark and a scheduler that learns from incomplete records while choosing individual tests. Its value must be demonstrated against strong alternatives; assembling the components alone is not sufficient novelty.

A focused literature review did not establish an exact matching TCAMS implementation. This is a proposed application and evaluation gap, not a verified first-ever architecture. Novelty and publication remain contingent on a deeper related-work review and the experimental results.

1. Available data and feasibility

The main source is the freely downloadable ChEMBL-NTD export of the GSK TCAMS collection. [1] The export was downloaded and audited on 27 September 2026. The paper reports 13,533 compounds; the current export contains **13,519 rows**. These are different counts and should not be substituted for each other.

Field	Role in this project
SMILES	Text representation of molecular structure; converted to fingerprints.
pXC50_3D7	Known parasite potency used to define the candidate pool. Higher values indicate greater potency.
PCT_INHB_DD2	Percentage inhibition in the Dd2 parasite assay; one result the scheduler may reveal.
PCT_INHIB_HEPG2	Percentage inhibition in a human-derived cell counterscreen; the other result it may reveal.

Preliminary audit stage	Count
Raw rows with all three measurements	13,187
Unique structures after largest-fragment canonicalization	13,461
Unique structures with all three measurements	13,129
Eligible at illustrative pXC50_3D7 >= 6.3	3,473
Also meeting Dd2 >= 80% and HepG2 <= 20%	1,224

Cleaning used the largest molecular fragment, canonical isomeric SMILES and median endpoint values for duplicate structures. No invalid SMILES were found. This is a preliminary audit, not a completed standardization protocol; charge, tautomer and duplicate handling must be frozen before evaluation. The eligible pool contains 1,996 scaffold keys, including an empty key for three acyclic compounds.

What these measurements support

Treat 3D7 potency as already measured: this is a follow-up confirmation problem after initial screening. The illustrative thresholds above establish feasibility only. Lock primary cutoffs before method comparison and report sensitivity to alternative cutoffs. They are computational definitions of a confirmed hit, not clinical recommendations.

Dd2 is a single-concentration result here, not a matched potency measurement. It does not support a resistant/sensitive potency ratio or a claim of resistance robustness. HepG2 is a cell-line counterscreen, not proof of human safety. TCAMS is enriched for initial hits, so findings apply to this selected pool, not arbitrary molecules.

Free data: Download the official TCAMS table. No purchased dataset or paid model API is needed.

2. Prediction architecture

A. Prepare a leakage-controlled replay

Keep a raw, versioned table and create a cleaned compound table. Group closely related structures by chemical scaffold before splitting. Reserve development families for tuning. In the final replay, start with a fully observed training seed, a separate calibration seed and a candidate pool from other families. All methods receive identical starting information.

Only the simulator can access unrevealed Dd2 and HepG2 values. The learning system sees structure, known 3D7 potency and an observation mask saying which follow-up results have been purchased. Missing measurements are never treated as failures. The main replay uses rows with both real follow-up outcomes available.

B. Encode each compound

Use RDKit Morgan fingerprints: a fixed-length numerical summary of structural patterns. A starting configuration is 2,048 bits plus standardized known 3D7 potency. Fit preprocessing only on permitted training information. Do not include IFI or other columns that summarize hidden assay history.

C. Learn the joint outcome

Train a small neural classifier, initially one hidden layer with 128 units, to estimate four probabilities: both checks fail; only HepG2 passes; only Dd2 passes; both pass. “HepG2 passes” means the inhibition value is below the predefined cutoff. A softmax makes the four probabilities add to one.

Learning from partial records: if both outcomes are known, train on their exact joint class. If only Dd2 is known, train on the sum of the two joint probabilities compatible with that outcome. Do the equivalent for HepG2-only records. A completely untested compound supplies no supervised outcome. This marginal-likelihood method is established statistical machinery, not a claimed invention.

This matters because the scheduler may stop testing a compound after one failure. Those incomplete records still teach the model about the measured endpoint. Their unmeasured outcomes remain unknown.

D. Check confidence and update after a result

Use a small ensemble, subject to runtime, and calibrate joint probabilities on the separate fully observed calibration seed. Check probability quality with Brier scores and reliability plots. Derive conditional probabilities after one test is revealed: the probability of passing the remaining test can change once the first outcome is known. [11,12]

Include independent fingerprint/random-forest models as a strong low-cost baseline. Compare the joint model with this simpler alternative. Calibration on an initial sample does not guarantee reliability after adaptive selection or chemical-family shift; measure that limitation rather than claiming guaranteed confidence.

3. Decision loop and budget

E. Choose the next compound and test

Let pD be the predicted chance of passing Dd2, pH the chance of passing HepG2, and pBoth the joint chance of passing both. Let cD and cH be their assumed costs. For an untested compound, compare these two routes:

Route	Expected cost until rejection or confirmation
Dd2 first; HepG2 only after a pass	cD + pD x cH
HepG2 first; Dd2 only after a pass	cH + pH x cD

Rank an untested compound using **pBoth / lower expected route cost**, and request the first test in that route. For a compound that has already passed one test, use its conditional chance of passing the remaining test divided by that test's cost. After a failure, stop spending on confirmation but retain its observed label for training.

This is a practical greedy scheduling rule, not a globally optimal policy. Add a small, prespecified exploration component, initially 10% of actions, that samples unmeasured tests across diverse compounds. Evaluate whether that addition helps. Uncertainty is a diagnostic and exploration signal; high confidence is not assumed to imply correctness.

F. Reveal, charge and retrain

The simulator reveals only the requested recorded result and subtracts its cost. A compound counts as confirmed only after both measured checks pass. Use small batches, initially 32 assay actions, then update the partial-observation model. Do not use outcomes from the same batch before the batch is completed. Do not request duplicate tests.

Continue until the fixed budget is exhausted. Report unfinished compounds with one result; they do not count as confirmed. A feasible action must fit the remaining budget. Freeze model settings and scheduling rules before the final replay.

G. Count all information costs

Report seed and calibration measurements explicitly. For example, 200 training compounds and 100 calibration compounds, each with both follow-up tests, cost 600 assay units when each test costs one. Show both total-budget curves and additional-budget curves. Known 3D7 potency is a declared sunk cost in this post-screening scenario.

A baseline the AI must beat

In the audited eligible pool, always purchasing both tests would reveal 6,946 results. With equal costs and stopping after a failure, Dd2-first would reveal 5,837; HepG2-first would reveal 5,230. These are full-pool arithmetic counts from existing outcomes, **not model performance**. They show why comparison only against "test everything" would exaggerate the AI contribution.

4. Evaluation and seven-week schedule

Primary endpoint: number of distinct, measured confirmed hits at matched total assay cost. Secondary endpoints: cost per hit, fraction of eligible hits recovered, confirmed scaffold diversity, probability calibration, unfinished tests and runtime. Use identical candidate pools, seeds and budgets across methods.

Comparison	Question it answers
Random and potency-ranked fixed routes	Does learning improve on simple prioritization?
Both fixed test orders with model ranking	Does choosing the test order actually help?
Compound-only active learning; purchase both tests	Does selecting individual tests add value?
Joint model with fixed routes	Does adaptive scheduling add value to the same predictor?
Adaptive route with complete-record-only learning	Does learning from partial records add value?
Full method with and without exploration	Is the exploration component useful?

Include an uncertainty-aware compound-ranking comparator inspired by recent multi-assay active learning. [8] Clearly label an adaptation rather than implying an exact reproduction. Run at least five replay seeds; report paired differences and variability, not only the best run. Vary assumed cost ratios (1:1, 1:3, 3:1) and a small prespecified threshold grid. These are sensitivity scenarios, not verified laboratory prices.

Scaffold separation tests the initial shift to unfamiliar families. After the model acquires labels from a family, later predictions in that family are no longer strictly unseen-family predictions. Report early and later rounds separately. Chronological or prospective claims are inappropriate without trustworthy timestamps or new experiments.

Week	Concrete output
1	Cleaned data, duplicate/scaffold audit and locked replay protocol.
2	Hidden-result simulator, cost accounting and fixed-route baselines.
3	Independent random forests and joint partial-observation predictor.
4	Conditional scheduler, confidence diagnostics and exploration variant.
5	Repeated matched-budget comparisons on final replay pools.
6	Ablations, threshold/cost sensitivity and error analysis.
7	Paper draft, figures, documented benchmark and reproducible code.

Deliverables: a versioned dataset-processing script; reusable assay-reveal simulator; benchmark splits and seeds; trained lightweight models; result curves; and a manuscript describing both gains and failures. No model comparisons have been run yet. The verified counts demonstrate data availability, not that the proposed method works.

5. Research references

The sources below establish the data and neighboring methods. They support a focused prior-work map; they do not prove that no matching study exists. Links lead to publisher, repository or paper pages.

[1] Gamo et al. (2010). Thousands of chemical starting points for antimalarial lead identification. Nature.
https://doi.org/10.1038/nature09107
TCAMS source and experimental context.

[2] Williams et al. (2015). Cheaper faster drug development validated by the repositioning of drugs against neglected tropical diseases. J. R. Soc. Interface.
https://doi.org/10.1098/rsif.2014.1289
Early active-learning precedent that includes malaria.

[3] Zhang et al. (2019). Bayesian semi-supervised learning for uncertainty-calibrated prediction of molecular properties and active learning. Chemical Science.
https://doi.org/10.1039/C9SC00616H
Uncertainty and active learning, including a malaria dataset.

[4] MAIP: a web service for predicting blood-stage malaria inhibitors (2021). Journal of Cheminformatics.
https://doi.org/10.1186/s13321-021-00487-2
Existing malaria prediction infrastructure.

[5] Tse et al. (2021). An Open Drug Discovery Competition: Experimental Validation of Predictive Models in a Series of Novel Antimalarials. J. Med. Chem.
https://doi.org/10.1021/acs.jmedchem.1c00313
Public antimalarial prediction and experimental evaluation.

[6] Godinez et al. (2022). Design of potent antimalarials with generative chemistry. Nature Machine Intelligence.
https://doi.org/10.1038/s42256-022-00448-w
Shows why malaria generation alone is not a new contribution.

[7] Lin et al. (2024). MalariaFlow: A comprehensive deep learning platform for multistage phenotypic antimalarial drug discovery. Eur. J. Med. Chem.
https://pubmed.ncbi.nlm.nih.gov/39173285/
Prior multistage antimalarial modeling.

[8] Shen et al. (2026). ChemScreener: an active learning enabled hit discovery workflow with WDR5 inhibitor case study. Journal of Cheminformatics.
https://doi.org/10.1186/s13321-026-01204-7
Recent multi-assay active-learning comparator.

[9] Gelbart, Snoek and Adams (2014). Bayesian Optimization with Unknown Constraints. UAI.
https://arxiv.org/abs/1403.5607
Constrained selection with separately evaluated outcomes.

[10] Hernandez-Lobato et al. (2016). A General Framework for Constrained Bayesian Optimization using Information-based Search. JMLR 17(160).
https://jmlr.csail.mit.edu/papers/v17/15-616.html
Decoupled evaluation is an established optimization idea.

[11] Hirschfeld et al. (2020). Uncertainty Quantification Using Neural Networks for Molecular Property Prediction. JCIM.
https://doi.org/10.1021/acs.jcim.0c00502
Why uncertainty estimates need empirical checks.

[12] Bai et al. (2025). Conformal Selection for Efficient and Accurate Compound Screening in Drug Discovery. JCIM.
https://doi.org/10.1021/acs.jcim.5c01633
Prior uncertainty-aware screening; assumptions matter.